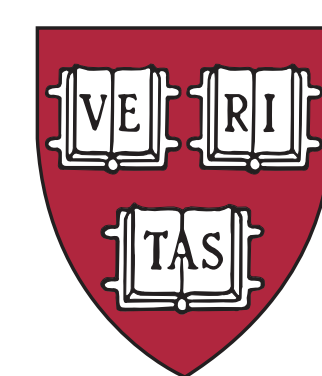




Multiplexing TMT and SILAC for accurate quantification of mammalian proteomes in MS2 mode

Nikolai Slavov^{†,✱}, Alexander van Oudenaarden[✱], and Bogdan Budnik[†]

[†] FAS Center for Systems Biology, Harvard University, Cambridge, MA 02138, USA

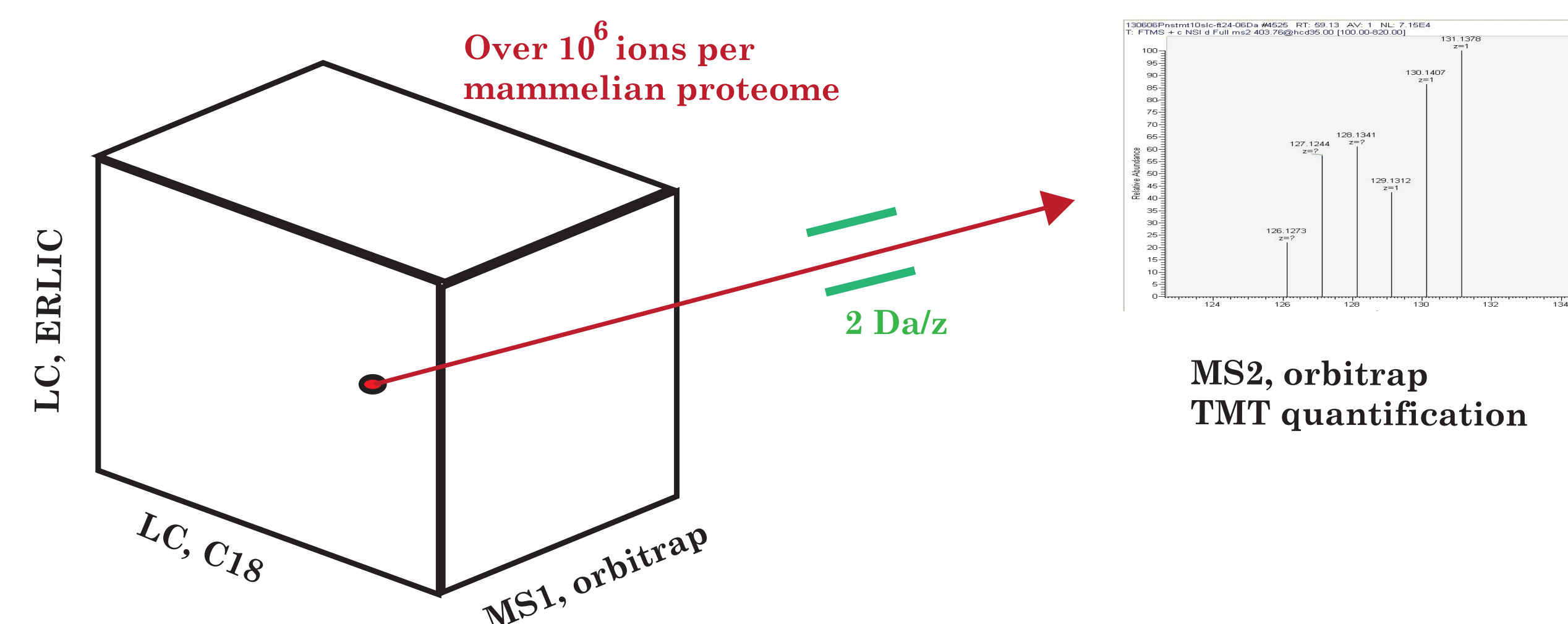
[✱] Hubrecht Institute, Uppsalalaan 8, 3584 CT, Utrecht, The Netherlands



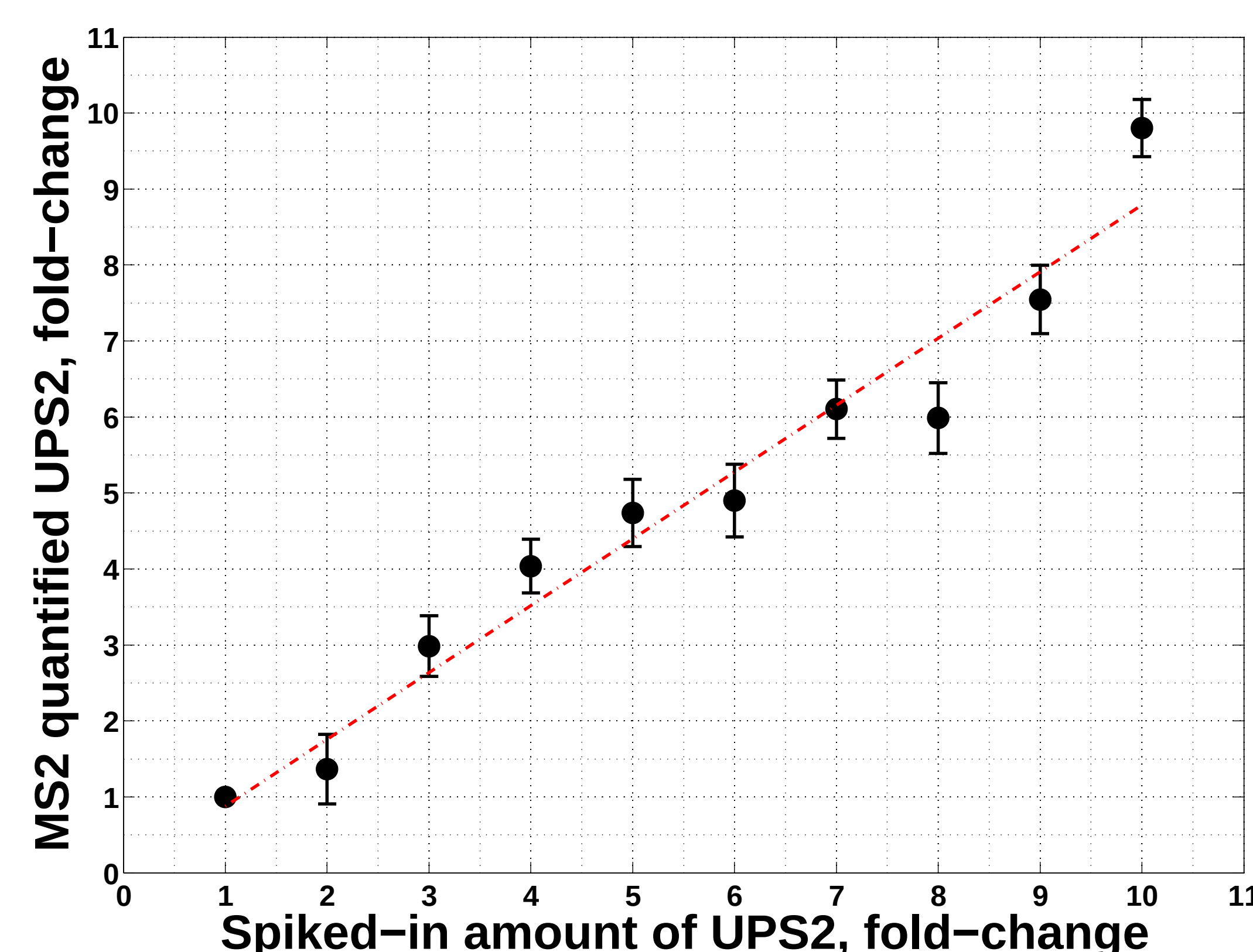
ABSTRACT

Despite the potential of Tandem Mass Tags (TMT) labeling to increase the throughput and accuracy of protein quantification, its accuracy and use remain limited by co-isolation interference. We developed experimental strategy that allows combining TMT and SILAC labeling for multiplicative increase in the number of samples that can be simultaneously quantified proteome-wide at MS2 level. We characterize the limits of TMT quantification and demonstrate comparable accuracy to that achieved by SILAC, with median fold-change error below 30 %. Combining SILAC and TMT labeling allows quantitation of whole organism proteomes with acceptable error and better coverage than TMT MS3-based quantification.

MINIMIZING CO-ISOLATION INTERFERENCE

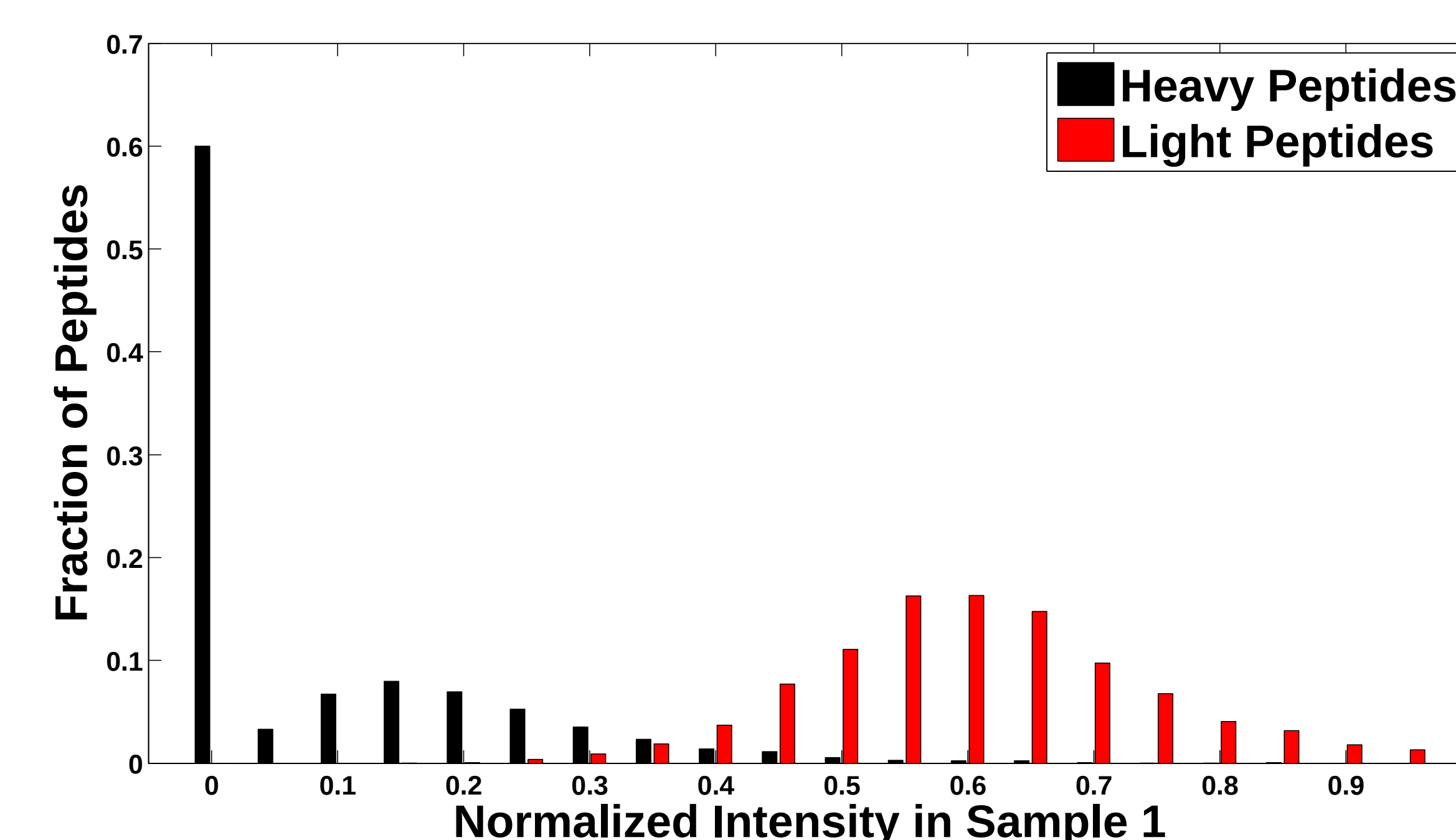


Overcoming challenges of protein quantification by isobaric tandem mass tags. Schematic diagram of the origin of co-isolation interference. Fold-changes estimated by MS are plotted against the fold-changes expected from the amount of universal proteomics standard (UPS2) added to sucrose fractions. All fold-changes are relative to the first data point, and the error bars are coefficients of variation. The fitted line has a slope of 0.88, indicating about 12% systematic underestimation (Slavov et al., 2014).

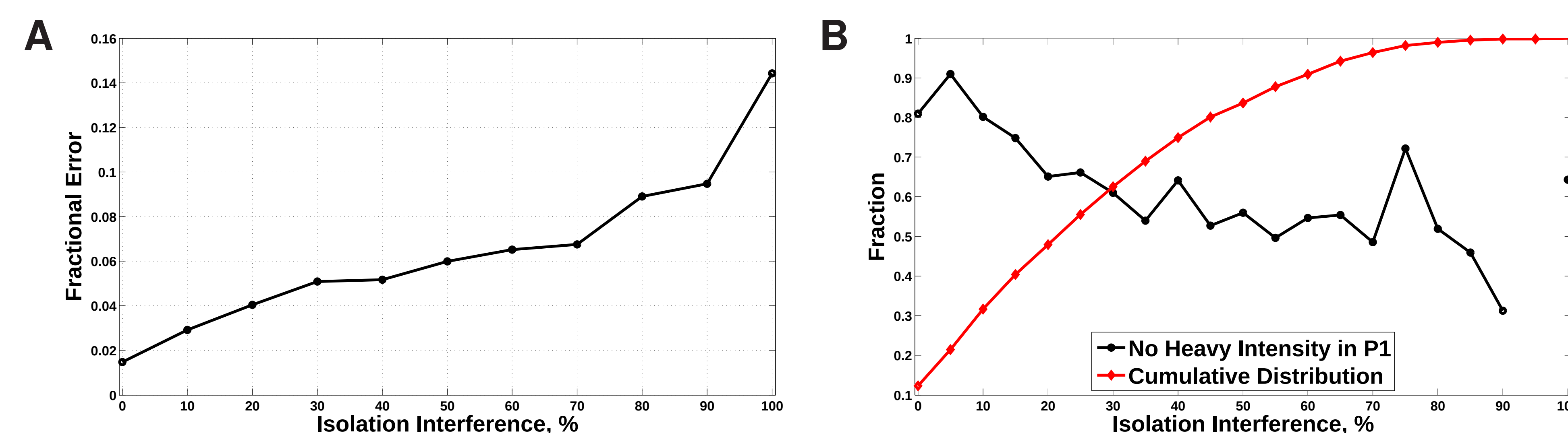


Accuracy of relative quantification in MS2 mode Fold-changes estimated by MS are plotted against the fold-changes expected from the amount of universal proteomics standard (UPS2) added to sucrose fractions. All fold-changes are relative to the first data point, and the error bars are coefficients of variation. The fitted line has a slope of 0.88, indicating about 12% systematic underestimation (Slavov et al., 2014).

COMBINING TMT AND SILAC



Low level of coisolated peptides Distribution of reporter ion intensities for heavy and light peptides in a sample containing only light proteins. Thus the signal for heavy peptides quantifies the magnitude of co-isolation interference.



Excluding peptides with high co-isolation interference improve the accuracy (A) Mean ratios of reporter ion intensities for the same heavy peptides between a sample collected before the introduction of heavy Lys and Arg and a sample collected 12 hours later. In the absence of co-isolation interference, this ratio should be zero. (B) Fraction of peptides having zero heavy intensity of the report ion in the channel having only light protein (collected before the introduction of heavy Lys and Arg) as a fraction of the co-isolation interference threshold imposed on the data.

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